MARKING KEY

2008/23051

Note: Specific examples have been chosen to illustrate responses. In many cases where the question allows for responses in various contexts, the example given reflects one specific context.

SECTION ONE: MULTIPLE-CHOICE [20 marks]

Question	Response
1	a
2	b
3	d
4	d
5	c
6	b
7	a
8	a
9	c
10	a
11	c
12	d
13	a
14	b
15	c
16	d
17	b
18	b
19	c
20	a

SECTION TWO: SHORT RESPONSE [70 marks]

Question 1 [12 marks]

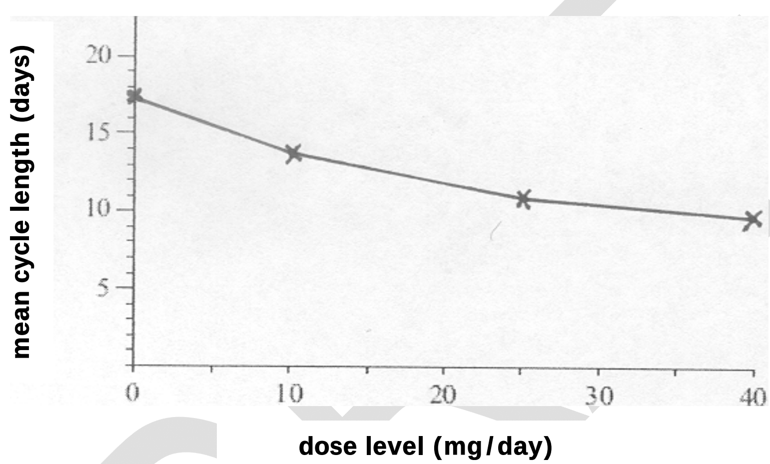
(a)

[4 marks]

<table>
<tr><th>Description</th><th>Marks</th></tr>
<tr><td>Progesterone dose level (mg/day)
<table>
<tr><td></td><td>0</td><td>10</td><td>25</td><td>40</td></tr>
<tr><td>MEAN</td><td>17.5</td><td>13.75</td><td>11.5</td><td>10.5</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td></tr>
</table>
</td><td>4
(1 mark for each mean calculated)</td></tr>
</table>

(b)

[1 mark]

<table>
<tr><th>Description</th><th>Marks</th></tr>
<tr><td><u>Effect of progesterone on length of the oestrous cycle</u>


</td><td>1</td></tr>
</table>

(c)

[4 marks]

<table>
<tr><th>Description</th><th>Marks</th></tr>
<tr><td>Any two of : replication, randomisation, control</td><td>2
(1 mark for each component)</td></tr>
<tr><td>How each minimises variation e.g.
• control, so that variations between treatments can be compared
• replication of each treatment, to overcome atypical responses exerting an influence due to very good or very poor environmental conditions</td><td>2
(1 mark for each explanation)</td></tr>
</table>

(d)

[1 mark]

Description	Marks
If the dose of progesterone increases, then the length of the oestrous cycle will decrease.	1

(e)

[2 marks]

Description	Marks
Conclusion 1: Progesterone injected does decrease the length of the oestrous cycle.	1
Conclusion 2: increasing the dose rate of progesterone does not significantly decrease the length of the oestrous cycle.	1

Question 2 **[12 Marks]**

(a)

[4 marks]

Description	Marks
• Metabolisable energy levels required by the animal are increasing before joining, • thus increasing the chance of conception • ME requirements increase due to growth of foetus, and • continues until weaning, when nutritional demands on mother decline.	1 1 1 1

(b)

[4 marks]

Description	Marks
State one management practice: e.g. improved nutrition	1
Reason: e.g. to meet the energy requirements of the animal	1
Timing: e.g. improved nutrition should occur prior to joining	1
Reason: e.g. to increase chances of conception	1

(c)

[4 marks]

Description	Marks
• Any two out of: genetics, temperature, day length, disease	2 (1 mark each)
• Associated explanation of each of the factors chosen e.g. genetics: selective breeding for fertility day length: the stimulus that controls breeding activity in seasonal breeders is the change in the ratio of daylight to hours of darkness.	2 (1 mark each explanation of 2 chosen factors)

Question 3 **[10 marks]**

(a)

[2 marks]

Description	Marks
• Ruminant • e.g. sheep, cattle, goats	1 1

(b)

[4 marks]

Description	Marks
• function of the rumen-reticulum is to break down the cellulose cell walls of plant materials for absorption • the rumen-reticulum contains a large population of microorganisms that • breakdown carbohydrates, involved in the synthesis of proteins and • vitamin B synthesis	1 1 1 1

(c)

[3 marks]

Description	Marks
• the gizzard is the unique feature • made up of muscular walls which breaks up the food mechanically • poultry	1 1 1

(d)

[1 mark]

Description	Marks
• an example of any farm monogastric: examples may include one of the following: pig, horse, dog, cat	1

Question 4 **[Total 12 Marks]**

(a)

[2 marks]

Description	Marks
Any two of the following: • cotton overalls buttoned to neck and wrists • washable hat • elbow-length PVC gloves	2 (1 mark each)

(b)

[6 marks]

Description	Marks
Step 1	
• weigh sheep	1
• calibrate applicator according to heaviest sheep in the race.	1
Step 2	
• apply chemical along the backline	1
• from the beginning of the neck to the tail head	1
Step 3	
• after all sheep have been treated, triple rinse equipment and	1
• dispose of rinsate in a disposal pit away from watercourses.	1

(c)

[4 marks]

Description	Marks
Two examples of incorrect chemical usage is identified e.g.	
• chemical applied inside withholding period	1
• no personal protective clothing	1
Consequence:	
• chemical levels in the animal product is above the maximum residue limit	1
• risk of skin contamination and thus poisoning	1
Note: these two examples are not exhaustive of all possibilities	

Question 5 **[Total 12 Marks]**

(a)

[1 mark]

Description	Marks
Name of farm product: e.g. MILK	1

(b)

[1 mark]

Description	Marks
one criterion is identified: e.g. The hygienic quality of fresh milk is tested by using the 'bulk milk cell count'.	1

(c)

[6 marks]

<table>
<tr><th>Description</th><th>Marks</th></tr>
<tr><td>Lists and describe any three viable actions.
Examples:
<table>
<tr><th>List</th><th>Reasons</th></tr>
<tr><td>Washing cows' udders</td><td>to reduce contamination of milk</td></tr>
<tr><td>Teat dipping with iodine solution</td><td>to reduce chance of bacterial infection (mastitis)</td></tr>
<tr><td>Thorough flushing and cleaning of dairy equipment and tanks</td><td>for hygiene reasons</td></tr>
<tr><td>Refrigeration of farm milk vats at 4 degrees C</td><td>to prevent bacterial growth</td></tr>
</table>
</td><td>6
(1 mark each for list and reason)</td></tr>
</table>

(d)

[4 marks]

Description	Marks
• Name of organisation: e.g. Dairy Corporation	1
• What they are responsible for: e.g. responsible for milk marketing	1
• What their aim is: e.g. to increase milk consumption and awareness	1
• How it is achieved: e.g. through campaigns such as ' Milk, Legendary Stuff '.	1

Question 6 **[12 marks]**

(a)

[6 marks]

Description	Marks
(Punnett square below)	
Genotype of cow: Mm	1
Phenotype of cow: Normal	1
Genotype of bull: Mm	1
Phenotype of bull: Normal	1
Percentage chance of calves with the disease: 25%	1
Percentage chance of calves without the disease: 75%	1

		BULL gametes	
		M	**m**
COW gametes	**m**	Mm	mm
	M	MM	Mm

(b)

[6 marks]

Description	Marks
• original bull (5 years earlier) must have been heterozygous for disease and passed on the recessive gene to the replacement cows	1
• 50% of replacement cows were heterozygous for disease	1
• new bull heterozygous for disease used over heterozygous cows	1
• probability of disease expressed in progeny is 25% which is approximately what was observed	1 1
• solution is to cull new bull and mate a homozygous unaffected bull with cows of unaffected phenotype.	1

SECTION THREE: PRODUCTION PRACTICES RESPONSE 20 marks

Question 7

(a)

[1 mark]

Description	Marks
state an enterprise and end product e.g. beef enterprise for the domestic market	1

(b)

[5 marks]

Description	Marks
Any five activities listed e.g. • Determine market specifications • Select and obtain the best animals to meet required market specifications • Nutrition and health management program required to reach goal end point • Comply with OSH and animal welfare guidelines • Comply with quality assurance • Select for market e.g. weight, condition score, fat score • Marketing process	5 (1 mark per relevant point made)

(c)

[4 marks]

Description	Marks
• list 2 items of machinery/equipment • reasons for efficiency of each item	2 (1 mark each) 2 (1 mark each)

(d)

[4 marks]

Description	Marks
• examples of costs and • examples of incomes • description of costs and incomes relevant to chosen enterprise	1 1 2

(e)

[6 marks]

Description	Marks
definition of sustainability i.e. statement incorporating • social, • economic and • environmental factors discussion of each aspect of sustainability (as above) with reference to chosen animal-product	 1 1 1 3

SECTION FOUR: EXTENDED RESPONSE **40 marks**

Question 8 *[20 marks]*

(a)

Description	Mark
• List 3 common methods available to prevent or control animal pests and diseases. The following examples could be listed: e.g. eradication, vaccination, chemical control, biological control, genetic control, management control, quarantine • Description of each of the 3 methods chosen.	3 (1 mark for each) 3 (1 mark for each)

(b)

Description	Mark
• Identifies one pest and host animal e.g. roundworm-barber's pole worm (pest) in sheep (host animal)	2
• provides a detailed explanation of at least two ▪ Symptoms e.g. paleness of skin, paleness of eye membranes, lack of stamina, loss of condition.	2
▪ Contraction i.e. (parasite life cycle) e.g. sheep ingest the infective larvae from pastures which develop into adult worms, these then produce eggs which are excreted onto the pastures, and the eggs hatch into larvae	2
▪ Consequences (Social, economic, environmental): discussion of how each of these factors affect the sustainability of the enterprise. e.g. resultant production losses through loss of condition/ weight of sheep reducing cents/kg, death in extreme circumstances which affects income and stock numbers.	3 (1 marks each for explanation of social, economic and environmental factors)
• Allocation of 5 marks for overall structure of essay ▪ introduction ▪ general flow of essay including good use of terminology ▪ conclusion	1 1–3 1

Question 9

[20 marks]

Description	Mark
• Identifies 3 breeding systems examples may include: random breeding, inbreeding, line breeding, cross breeding	3
• Describes each of the three processes e.g. line breeding is where breeding is based on a single common ancestor (a sire or dam) used over several generations of mating.	6 (2 marks for each description)
• Describes at least 2 points of how each of the three breeding systems improve animal production e.g. value of line breeding depends on the degree of superiority of the outstanding individual used. A high degree of uniformity of type and production is obtained.	6 (2 marks for each point of description)
Allocation of 5 marks for overall structure of essay • introduction • general flow of essay including good use of terminology • conclusion	 1 1–3 1

Question 10

[20 marks]

Description	Mark
• Identifies, and provides the characteristics of three factors that impact on profitability	6
• Outlines 1 management decisions for each factor to maintain profitability	3 (1 per factor)
• Provides detailed relationships between the farm mangers decisions and each factor to maintain profitability	3
• Links each decision long term sustainability.	3
Allocation of 5 marks for overall structure of essay	
• introduction	1
• general flow of essay including good use of terminology	1–3
• conclusion	1

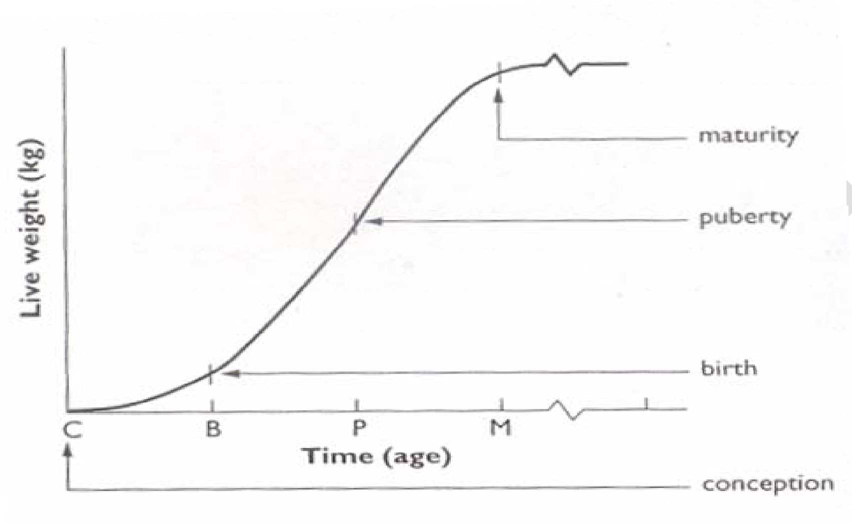
Question 11

[20 marks]

(a)

<table>
<tr><th>Description</th><th>Mark</th></tr>
<tr><td>

<u>Generalised growth curve of an animal</u>



The following labels included in diagram
A: Y axis – Live weight (kg) or (g)
B: X axis – Time and/or age
C: Birth
D: Puberty

</td><td>4
(1 mark for each)</td></tr>
<tr><td>Details of the shape of the curve:
• during early pregnancy growth rate of foetus is slow and increases in the last one-third of the pregnancy</td><td>1</td></tr>
<tr><td>• after birth, growth rate continues to increase until it reaches a maximum around the time of puberty (point of inflection)</td><td>1</td></tr>
<tr><td>• from puberty, growth rate slows as the animal reaches its mature size.</td><td>1</td></tr>
</table>

[Graph from: Brown, L., Hindmarsh, R., & McGregor, R. (2001). *Dynamic agriculture: Book 3* (2nd ed.). Sydney: McGraw-Hill]

(b)

• Definition of genetic potential: e.g. Genotype of identified animal sets the upper limit to its possible growth rate and development. If the environment is non-limiting, the animal will reach its genetic potential for growth and development. • Three environmental factors: e.g. any three of nutrition, climate, disease, stress	2 3 (1 mark for each
Management of any of the three factors: **Nutrition:** this refers to the amount and quality of feed available. Supplements can be given when nutrition is inadequate **Climate:** extremes in climate may affect growth. In very hot weather (animal may stop grazing) shade should be provided. In very cold weather (energy lost in keeping warm) shelter should be provided **Disease:** may stunt growth. Prevented or controlled by vaccination or drenching **Stress:** reduced by adjusting the stocking rate and using appropriate husbandry practices.	3 (1 mark for each)
Allocation of 5 marks for overall structure of essay • introduction • general flow of essay including good use of terminology • conclusion	 1 1–3 1

EXAM QUESTIONS
MAPPING TO COURSE CONTENT

Animal Production Systems
Stage 2
Relationships between content, and exam questions

Stage 2			
Unit 2A	**Questions**	**Unit 2B**	**Questions**
KNOWLEDGE FOUNDATIONS FOR ANIMAL PRODUCTION SYSTEMS		**KNOWLEDGE FOUNDATIONS FOR ANIMAL PRODUCTION SYSTEMS**	
Systems ecology • impact of animal production systems on natural resources e.g. soils, water, air, biodiversity, natural ecosystems • impact of natural resources on animal production systems.	**MC2**	**Systems ecology** • structure and function of systems i.e. material and information flows and feedback loops.	**MC8, MC19**
Animal anatomy and physiology • physiology of digestion, reproduction, respiration, circulation • cell structure and function of organelles • cell division for growth and reproduction • conception, pregnancy, birth, lactation • breeding cycles (monoestrous, polyoestrous, seasonally polyoestrous).	**Q1, Q2, Q3** **MC6, MC7** **MC4, MC17, Q11** **MC13**	**Animal anatomy and physiology** • endocrine system and the role in breeding behaviour and reproduction • variations in quality and causes e.g. breed, nutrition, pre- and post-harvest conditions • response curves and the law of diminishing returns.	**Q1** **Q5** **Q11**
Animal nutrition • feed intake and feed conversion ratios • feed on offer and stocking rates.	**Q10 Q11**	**Animal nutrition** • feed requirements for maintenance, growth and reproduction • ration composition to produce to market specifications • rates of growth and nutrient requirements.	**Q2 Q11** **Q11**
Animal health • impact of pests and diseases on production • life cycles of pest and disease organisms i.e. external and internal • prevention and treatment of pests and diseases e.g. vaccination, mulesing drenching, jetting, • prevention of spread of pests and diseases i.e. biosecurity, quarantine, eradication.	**Q4 Q8**	**Animal health** • pest and disease control systems i.e. cultural, chemical, biological, and integration of methods (IPM) • immunity resulting from vaccination i.e. antibodies and antigens.	**Q4** **Q8**

Stage 2			
Unit 2A	**Questions**	**Unit 2B**	**Questions**
KNOWLEDGE FOUNDATIONS FOR ANIMAL PRODUCTION SYSTEMS		**KNOWLEDGE FOUNDATIONS FOR ANIMAL PRODUCTION SYSTEMS**	
Animal improvement • aims of breeding and selection i.e. productivity, quality, adaptation to environments, fecundity, disease resistance • sources of variation and methods used for selection • selection criteria within a breed i.e. quantitative characteristics, heritability.		**Animal improvement** • gamete production i.e. meiosis • genetic principles i.e. genes, alleles; dominant, recessive, genotype, phenotype, inheritance • genotype by environmental interaction i.e. genetic gain • breeding systems i.e. inbreeding, line breeding, crossbreeding, genetic engineering.	**MC15** **MC5** **Q11** **Q9**
Economics and markets • animals and enterprises to meet market specifications • budgets i.e. gross margins, partial and whole farm (541) • marginal costs and marginal returns (524) and the law of diminishing returns (517–8).	**Q5**	**Economics and markets** • elasticity of supply and demand (535) • factors affecting supply and demand (536) • marketing systems • quality assurance e.g. ISO 1400, HACCP.	**Q5 Q7 Q5**
Requirements for sustainable production • conservation of biodiversity and functioning natural ecosystems • maintenance and improvement of soil quality, water quality, air quality • stewardship of natural and farming resources including technologies • prevention of the spread of pests and diseases i.e. biosecurity, quarantine, eradication.		**Requirements for sustainable production** • sustainability of current management practices i.e. balance of short-term financial needs with long-term maintenance and improvement of resources • restoration of degraded natural resources • government legislation for the protection of the environment • safe and efficient production, transport, storage and marketing of agricultural produce.	**Q7**
Improving knowledge, skills and practices through investigation • participation in investigations using the scientific method i.e. formulate and test an hypothesis and report outcomes e.g. breeds and feed trials • impact of current research and promotional literature.	**Q1**	**Improving knowledge, skills and practices through investigation** • data handling i.e. tables, graphs, means and levels of statistical significance • conclusions from own or others investigations and recommendations based on findings.	**MC11, Q1** **MC9, MC12, MC15, MC20, Q1**

Stage 2			
Unit 2A	**Questions**	**Unit 2B**	**Questions**
MANAGEMENT OF SELECTED ENTERPRISES FOR SUSTAINABILITY		**MANAGEMENT OF SELECTED ENTERPRISES FOR SUSTAINABILITY**	
Select and produce ready for market • types of land and labour used in enterprises • stocking rates used in production systems • calendars of operations for production cycles • monitoring the environment e.g. weather, temperature, humidity • selecting, preparing and applying registered pesticides to label specifications and cleaning up after use. • selecting and using resources and equipment for animal handling and production.	**MC10, MC14, MC16, Q4**	**Select and produce ready for market** • management of off-farm threats to natural and farm resources e.g. quarantine and biodiversity • management of the growth and development of animals to maximise productivity • identifying and monitoring bio-security requirements. • skills in practical animal management during routine operations according to OHS and animal welfare requirements.	**Q7** **Q11** **MC1, MC2, MC3**
Budgets and finance • role of budgets and identification of items most likely to impact on profit • interpretation of supply and demand information for a product.		**Budgets and finance** • sales reports to plan production and marketing • financial records to guide decision-making.	**MC17, MC18, Q7**
Animal nutrition management • calculate feed on offer and adjust stocking rates • calculate feed intake to obtain feed conversion ratio • compare feed conversion ratios to industry benchmarks.		**Animal nutrition management** • estimate feed requirements for maintenance, growth and reproduction • calculate and deliver feed rations to produce to market specifications • assessing quality of product against market requirements.	**Q11**
Animal health management of pests and diseases • management programs for pests and diseases i.e. integrated pest management (IPM) over several years and across enterprises • selection, preparation and application of registered veterinary chemicals to label specifications and cleaning after use.	**MC10, Q4**	**Animal health management of pests and diseases** • population dynamics and growth of pests and diseases • tactics and strategies for the regulation of population dynamics • population dynamics in response to tactical controls.	